Phosphatase activity assay with bis-pNPP and specific activity

What you're going to get / What you expect

The specific activity (U/mg) of each purified enzyme at process conditions (pH 5.0, 30 ºC). bis(p-nitrophenyl) phosphate is a phosphodiester that mimics the bulky, stable organophosphates of a real effluent better than pNPP. Its hydrolysis releases p-nitrophenol (pNP), a chromophore read at 405 nm. One unit (U) is the amount of enzyme that releases 1 umol of pNP per minute under these conditions.

Materials list (per kinetic series)

- bis(p-nitrophenyl) phosphate (bis-pNPP), 25 mM.
- Citrate buffer 50 mM, pH 5.0.
- 1 M NaOH (to stop the reaction and develop the colour).
- Purified enzyme; distilled water (spontaneous-hydrolysis control).
- 96-well plate; multimode microplate reader (405 nm); temper block at 30 ºC.

Steps

1. Prepare the substrate mixture in an Eppendorf tube: 32 uL of 25 mM bis-pNPP plus 688 uL of 50 mM citrate buffer pH 5.0. Citrate is used because the real degumming effluent is rich in citric acid and its salts.
2. Temper the tubes for 5 minutes at 30 ºC.
3. Start the reaction by adding 80 uL of enzyme (or distilled water for the spontaneous-hydrolysis control), keeping a constant 9:1 substrate-to-enzyme ratio and 30 ºC, and start the timer.
4. At 0, 2, 5, 10, 15, 20 and 30 min, transfer 100 uL of the reaction to a well pre-loaded with 100 uL of 1 M NaOH; this stops the reaction and develops the pNP colour by raising the pH.
5. Prepare the t = 0 point directly in the well, combining 90 uL of substrate mixture with 10 uL of enzyme (same enzyme amount as the other aliquots).
6. Read the whole plate at 405 nm and subtract the enzyme-free well at each time point to correct for basal autohydrolysis.
7. Convert net absorbance to pNP concentration and compute the specific activity with the equations below (take the initial rate as the slope between t = 0 and t = 1 min, where the rate is constant).

Conversion of absorbance to pNP concentration (Lambert-Beer):

$$[pNP]\ (uM) = (A405(sample) - A405(blank)) / (e \cdot l)$$

$$e = 0.0185\ uM^{-1}\ cm^{-1};\ l = 0.56\ cm$$

Specific activity:

$$m = ([pNP] \text{ at } t=1 \text{ min} - [pNP] \text{ at } t=0) / (1 \text{ min}) \quad (\text{umol } L^{-1} \text{ min}^{-1})$$

$$\text{Specific activity (U/mg)} = (m \cdot 0.0002 \text{ L}) / (c \cdot 0.01 \text{ mL})$$

$$c = \text{protein concentration by Bradford (mg/mL)} ; \; 1 \text{ U} = 1 \text{ umol pNP min}^{-1}$$

Troubleshooting

- Raw product curves can mislead: normalise by protein before ranking enzymes (the raw order can reverse after normalisation).
- pNP is only coloured at basic pH, so this is an end-point assay: do not try to follow it continuously at pH 5.0.
- Use citrate buffer, not phosphate, to match the composition of the real effluent.